QUESTION No- I (A) - Preparation and Demonstration of Slides	
Note: Teacher has to prepare a temporary slide using fresh specimen for demonstration. (During examination permanent slides can be used if temporary slide preparation is not possible).	
Exercise 1	T.S. of Mature anther
Exercise 2	L.S. of an Angiospermic ovule
Exercise 3	T.S. of *Nerium* leaf
QUESTION No- II (B) - Fresh or preserved specimens	
Exercise 4	Natural methods of vegetative propagation in plants - Rhizome, Sucker, Epiphyllous buds.
Exercise 5	Adaptations of flowers for pollination by different agents – Wind, Insects.
Exercise 6	Structure of Dicotyledonous seed – Gram (*Cicer*).
Exercise 7	Dispersal of seeds by various agents – Wind, Water, Animal.
Exercise 8	Ecological adaptations of plants - Hydrophytic, Xerophytic, Halophytic and Epiphytic.
QUESTION No- III (C)- Models / Photographs / Charts	
Exercise 9	Types of ovules – Anatropous, Orthotropous, Campylotropous
Exercise 10	Picture of a vector (pBR 322)
Exercise 11	Plant tissue culture – Callus with plantlets
Exercise 12	Types of ecological pyramids – Number, Biomass, Energy
QUESTION No- IV (D) - Problems – Genetics and Ecology	
Exercise 13	To verify Mendel's Monohybrid cross
Exercise 14	Analysis of seed sample to study Mendelian Dihybrid Ratio
Exercise 15	Flow of energy and Ten percent law
Exercise 16	Determination of population density and percentage frequency of different plant species of given area by Quadrat method
Exercise 17	Chromosomal aberration – Deletion, Duplication, Inversion
Exercise 18	Genetic / Linkage maps
QUESTION No- V (E) - Experiments	
Exercise 19	Dissect and display the Pollinia of *Calotropis*
Exercise 20	Study of pollen germination on a slide
Exercise 21	Study of pH of different types of soils
Exercise 22	Water holding capacity of garden soil and road side soil
Exercise 23	Isolation of DNA from plant material
QUESTION No- VI (F) -Economic importance of plants	
Exercise 24	Economically important plants and their uses Wheat, Black pepper, Cotton, Keezhanelli, Green gram, Banana
Exercise 25	Economically important plant products and their uses:Sesame / Gingelly oil, Rubber, Aval (Flaked rice), Rose water, Henna powder,Aloe gel

TN_GOVT_XII_BOTANY_PRACTICAL_MANUAL_PAGES_284-312.indd 272 3/8/2020 3:05:51 PM